

Darren Wang

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EDUCATION

University of Pennsylvania

B.S.E. in Computer Science; GPA: 3.99/4.0

May 2028

Philadelphia, PA

- **Coursework:** Operating Systems, Databases, Algorithms, Scalable Cloud Computing, Machine Learning.

TECHNICAL SKILLS

Languages: Python, C, C++, Java, OCaml, SQL, JavaScript, TypeScript, HTML, CSS

Frameworks & Tools: React, Next.js, Tailwind CSS, Node.js, Express, PostgreSQL, PyTorch, scikit-learn, NumPy, pandas, Polars, Docker, AWS, Apache Spark

EXPERIENCE

Computational Social Science Lab, University of Pennsylvania

June 2026 – Aug. 2026

Research Assistant – Software Engineering

Philadelphia, PA

- Merged a NOMAD optimization using spatial candidate search and temporal blocking, accelerating radius-based contact estimation by 35.6× (95.7s to 2.69s) on a 500-user benchmark while preserving identical output.
- Developed and deployed a Next.js and ECharts experimentation platform to evaluate 8 stop-detection algorithms on randomized synthetic GPS trajectories, featuring interactive parameter sweeps and synchronized geospatial and timeline playback for a NetMob submission.

University of Pittsburgh

Aug. 2023 – Present

Research Software Engineer – Model Evaluation

Pittsburgh, PA

- Published first-author research in *Journal of Computational Chemistry* on an AlphaFold 2, ZDOCK, and ANI-2x docking protocol evaluated across 62 protein-peptide systems, achieving 34% Top-1 and 45% Top-3 success.
- Built reproducible AlphaFold 3 evaluation pipelines for 2,000+ wild-type, mutant, and chimeric predictions, using mutation and ligand-swapping experiments to analyze structural robustness and confidence against experimental $\Delta\Delta G$.
- Built Boltz-2 classification and regression pipelines from 1,024-dimensional neural representations, scalar outputs, and docking features, using grouped cross-validation to prevent related ligand variants from leaking across folds.

University of Pennsylvania

Sept. 2026 – Present

Teaching Assistant – Algorithms and Database and Information Systems

Philadelphia, PA

- Support 350 students across Algorithms and Database and Information Systems through office hours and Ed Discussion, covering algorithms, debugging, relational modeling, and SQL.

PROJECTS

PennOS | C, Operating Systems, File Systems

May 2026

- Built a user-level Unix-like operating system in C that multiplexes thread-backed processes through 100 ms preemption and provides an interactive shell with redirection, executable scripts, and foreground/background job control.
- Designed a weighted-credit scheduler with round-robin fairness across three priority classes, measuring a 9.00:5.97:4.00 CPU split against its 9:6:4 target over 1,252 quanta; implemented wait/signal semantics, zombie reaping, and orphan reparenting.
- Engineered a persistent FAT-style file system with memory-mapped allocation metadata and chained-block storage, adding reference-counted descriptor inheritance, permissions, and exclusive-writer coordination across processes.

Anime Recommendation Query Engine | React, Node.js, PostgreSQL, AWS RDS

May 2026

- Built a React/Express recommendation and analytics app over 31M+ user-anime interactions, backed by PostgreSQL on AWS RDS with search, filtering, and co-viewer queries.
- Reduced representative recommendation latency from ~35s to ~50ms by profiling slow joins and applying indexes, materialized views, query restructuring, and Express API caching.

Distributed Social Graph Engine | Apache Spark, AWS EMR, DynamoDB

Dec. 2025

- Implemented weighted graph adsorption on Spark/EMR over friendship, interest, and article-engagement data from DynamoDB, materializing personalized news recommendations through iterative label propagation.
- Decoupled cluster computation from interactive requests by submitting jobs through Apache Livy, polling completion asynchronously, and refreshing feeds after Spark persisted recommendation results.

Concurrent HTTP Search Server | C++, Multithreading, Socket Programming

Dec. 2025

- Built a multithreaded C++ HTTP search server with a worker thread pool and non-blocking accept loop, serving concurrent file and query requests without spawning a thread per connection.
- Load-tested the server, measuring 8.0× throughput scaling from 1 to 8 concurrent clients across its 8-worker pool and ~8,900 requests/second at saturation with p95 latency under 9 ms.